Next-30-minute rainfall exceedance probability: C report

Experiment C_v1_20261012; B_v2 features; data v1/rainfall_v1_r3. This is a C chapter; the integrated report and member details are in submission.

Results

Prespecified equal-threshold/equal-year Brier selected Decision tree: development mean 0.017585. After selection, refit on a fixed 10% 2017–2023 anchor sample and evaluate all supported 2024 anchors once: mean Brier 0.017736, training-frequency baseline 0.028883, skill 38.59%. Positive skill means improvement, within the documented data/eligibility scope.

Truth, time and supported inputs

Predict P(next-30-minute total > threshold). A uses Decimal(38,18) for the six exact endpoints t+300,...,t+1800; equality is negative, unknown future is not negative. Supervision filters the correct role, full past/future windows and actual temporal boundary, including endpoint equality to the next block start. Internal training-year crossings remain usable. Past history requires 12 exact positions t–3300,...,t. Current interval (t–5min,t] is assumed available at t; official interval meaning and publication delay remain unverified.

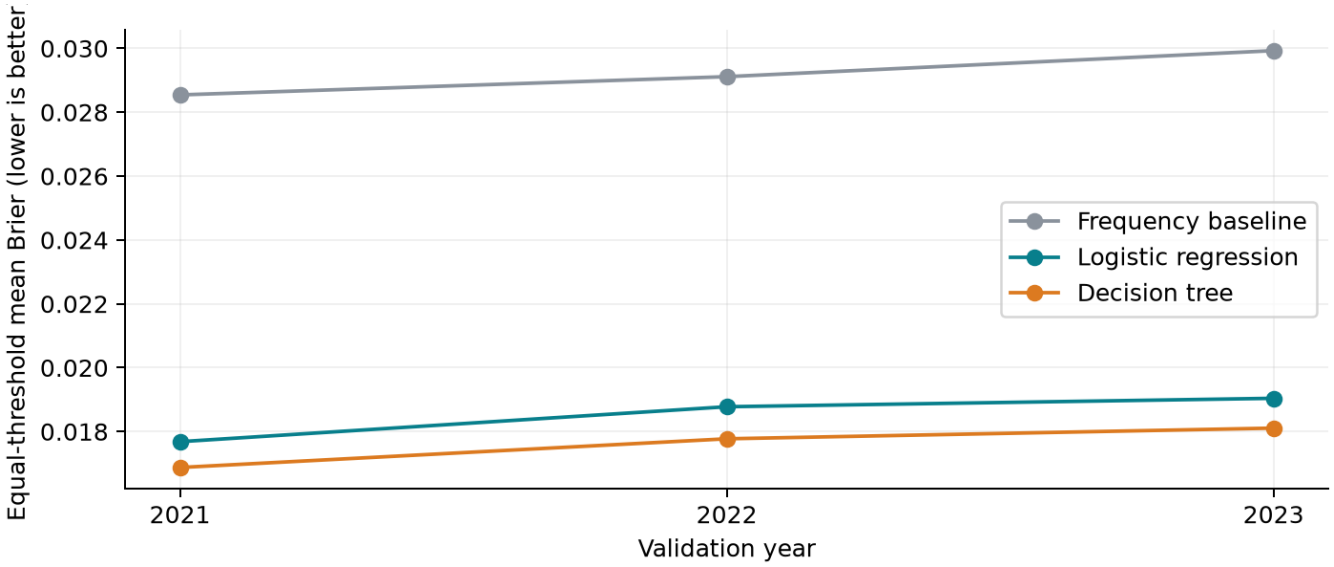
Use B 26 past-only numeric features, numeric threshold and training-only station one-hot coding. Future truth/completeness/eligibility/role are excluded. Inference checks only existing history, supported station, grid and threshold. Support is frozen from earliest full 2017–2020 training: at least 100 positives/negatives per candidate. Thresholds 0,0.5,1,2,5 mm all pass; arbitrary continuous-threshold generalization is not claimed.

E2 and model selection

Expanding folds train 2017–2020/validate 2021, 2017–2021/2022 and 2017–2022/2023; lock then refit 2017–2023/test 2024. A 1% pilot checks implementation; formal training samples 10%, evaluation is full eligible scope. pmod(xxhash64(station_id,prediction_epoch_s,seed),10000), seed=20261012, samples anchors before all threshold versions. Sampling is independent of outcomes with no class reweighting. Fit station StringIndexer/OneHotEncoder and LR standardization on each training fold only.

Candidates: per-threshold training frequency; L2 LR regParam=0.01/maxIter=40/tol=1e-6; tree maxDepth=5/maxBins=32/minInstancesPerNode=100 expanded rows. Average Brier equally over thresholds, then years. Tie tolerance 1e-5 prioritizes baseline, LR, tree; choices were fixed before formal scores. This limited candidate set is not exhaustive tuning.

Model	2021	2022	2023	Mean	Year range
Frequency baseline	0.028542	0.029113	0.029928	0.029194	0.001386
Logistic regression	0.017683	0.018778	0.019041	0.018501	0.001359
Decision tree	0.016875	0.017773	0.018108	0.017585	0.001233



Rolling Brier

selection.json was persisted before 2024 scoring; test metrics did not guide selection.

Actual input joins and scoring coverage

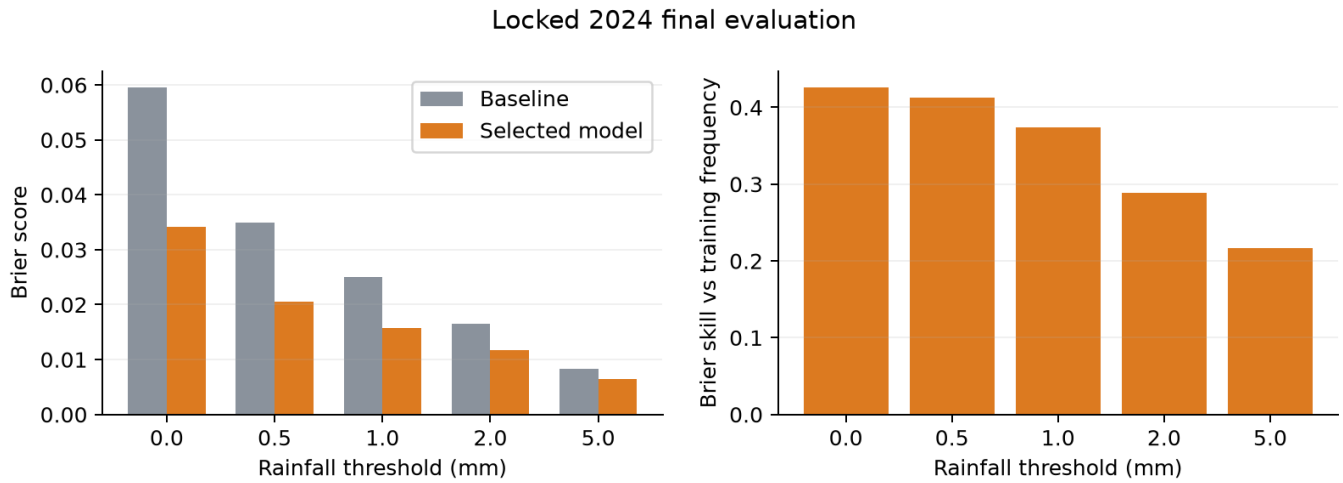
Each role/eligibility/station-filtered and training-sampled anchor set is audited before threshold expansion with B_join_audit.audit_anchor_join. Require unique/null-free keys, every left anchor matched and no row inflation; unused right-side features are allowed. All checks passed. Expanded station/epoch/threshold keys are unique, each anchor has five nonnull labels. Eligibility and role never enter model inputs.

Stage	Train anchors	Observed eval anchors	Eligible role anchors	Unknown-station exclusion	Scored anchors	Observed coverage
fold1	1,995,803	7,044,646	6,976,044	0	6,976,044	99.026%
fold2	2,693,419	7,133,230	7,072,758	0	7,072,758	99.152%
fold3	3,401,643	6,742,890	6,686,625	0	6,686,625	99.166%
final	4,071,679	6,406,260	6,338,426	0	6,338,426	98.941%

Coverage denominator is observed evaluation anchors, not an ideal complete station calendar. History/future exclusions overlap. Separate coverage.json retains individual history/joint/boundary counts. The deployed model rejects new stations.

Locked 2024 final evaluation

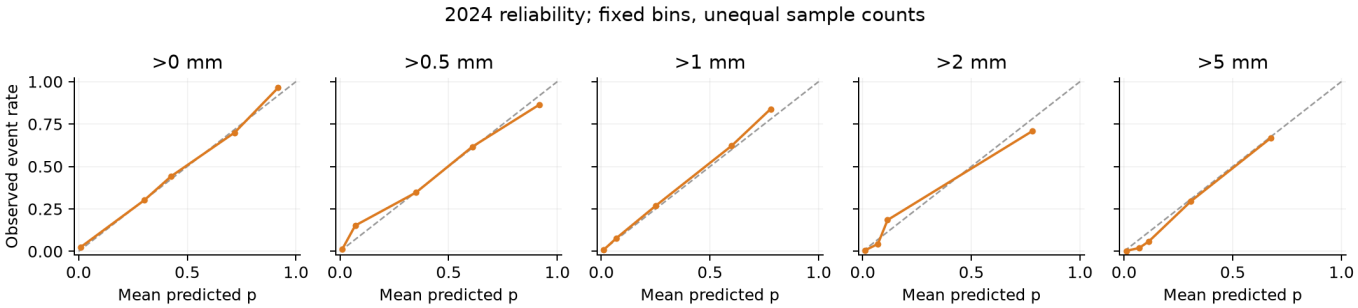
Threshold/mm	n	Positive rate	Baseline Brier	Selected Brier	Skill	Log loss	PR-AUC	ROC-AUC
0	6,338,426	6.361%	0.059571	0.034183	42.62%	0.150547	0.681425	0.805980
0.5	6,338,426	3.627%	0.034960	0.020544	41.24%	0.090920	0.610258	0.839476
1	6,338,426	2.571%	0.025050	0.015685	37.38%	0.070748	0.550593	0.839012
2	6,338,426	1.684%	0.016555	0.011781	28.83%	0.055588	0.438596	0.831617
5	6,338,426	0.835%	0.008279	0.006486	21.66%	0.036065	0.357150	0.831203



Final Brier

Overall Brier gives equal weight to five threshold scores; skill=1−model Brier/baseline Brier. Baseline is refit on the same final sampled training. Spark BinaryClassificationEvaluator uses numBins=1000 for approximate AUC; single-class groups are NA. Log loss clips only for numerical stability to [1e-15,1−1e-15]; saved raw probabilities are unchanged.

Reliability and limitations



Final reliability

Bins were fixed in advance and have unequal support; counts are in final/decision_tree_calibration.csv. No test calibration was applied. Adjacent-threshold probability increases in the test: 0 pairs. Raw predictions were not corrected for monotonicity; future inputs are not guaranteed monotone. Rare higher thresholds require support-aware interpretation rather than comparing raw Brier alone.

Station/weather composition changes, missing data and overlapping windows limit generalization. No independent-sample confidence intervals were computed. Only 10% training anchors and few candidates were evaluated, without full-training comparisons, feature ablation or external weather/radar inputs. Reject incomplete history/unknown stations. Three development years and one test year do not establish climate trends or live-deployment performance.

Model persistence and prediction

The complete PipelineModel and prediction_contract.json in trained_model/ preserve station encoding. A fresh Spark process compared 100 test anchors/500 probabilities: maximum absolute reload error 0.000000. Past-only single-anchor predictions matched offline results at all five thresholds. Unknown station, unsupported threshold, missing history and off-grid time were correctly rejected. C_predict.RainfallPredictor.predict accepts observations, station, endpoint epoch and threshold; output includes those identifiers and probability. No future label is required to accept a request.

Training impurity importance is diagnostic, not causal or independent ablation evidence. Full weights/tree rules are in final/.

Feature	Normalized impurity importance
rain_sum_15m_mm	0.616424
threshold_mm	0.212747
current_5m_mm	0.108971
rain_sum_30m_mm	0.048103
minutes_since_last_rain_60m_capped	0.006784
consecutive_wet_slots_60m	0.004353
rain_change_15m_mm	0.002619

Computing environment and timing

Item	Measured/configured
Machine	Apple M3 Max
Physical memory	36 GB
OS	macOS-14.1-arm64-arm-64bit
Python/Spark/Java	3.11.17 / 3.5.7 / 17.0.20.1
Spark	local[2], 6g heap, 128 shuffle partitions, 32 MiB reads, AQE on/coalescing off
Storage	Stage anchor/prediction Parquet, DISK_ONLY projected training vectors, no full observation cache
Peak memory	Not measured

Stage	Seconds
fold1	346.4
fold2	383.9
fold3	412.3
final	312.1

Stage elapsed includes actual joins, writes, expansion checks, preprocessing, fitting, full scoring and metrics/diagnostics. Excludes prior input hash verification, Spark startup, figures/report and later packaging/reload. Candidate fit_seconds measures classifier fit only, excluding shared preprocessing and frequency estimation. Single local measurements do not control OS file cache and are not stable benchmarks.

AI use and submission scope

OpenAI Codex assisted protocol implementation, code, execution, verification, figures and English drafting. Scores are actual saved Spark results; automated checks do not replace group understanding/review. The integrated submission contains A/B/C chapters and member identities; Canvas upload remains a user action. References: Spark 3.5.7 classification/regression and ML Pipelines documentation; archive/history_v1/docs/E2_ROLLING_VALIDATION.md.

Appendix: 2021 complete development metrics

Model	Threshold	n	Positive rate	Brier	Log loss	PR-AUC	ROC-AUC
Frequency baseline	0	6976044	0.060931	0.057228	0.229608	0.060931	0.500000
Frequency baseline	0.5	6976044	0.035763	0.034500	0.154486	0.035763	0.500000
Frequency baseline	1	6976044	0.025885	0.025233	0.120534	0.025885	0.500000
Frequency baseline	2	6976044	0.017239	0.016950	0.087364	0.017239	0.500000
Frequency baseline	5	6976044	0.008877	0.008800	0.050912	0.008877	0.500000
Logistic regression	0	6976044	0.060931	0.034720	0.139207	0.669028	0.875574
Logistic regression	0.5	6976044	0.035763	0.019294	0.083506	0.640883	0.894947
Logistic regression	1	6976044	0.025885	0.015335	0.068513	0.585358	0.890219
Logistic regression	2	6976044	0.017239	0.012055	0.054929	0.507219	0.881205
Logistic regression	5	6976044	0.008877	0.007010	0.034073	0.405241	0.878675
Decision tree	0	6976044	0.060931	0.031502	0.141514	0.703433	0.807009
Decision tree	0.5	6976044	0.035763	0.019297	0.086308	0.639686	0.843876
Decision tree	1	6976044	0.025885	0.015246	0.069204	0.581411	0.841533
Decision tree	2	6976044	0.017239	0.011711	0.055536	0.468497	0.830913
Decision tree	5	6976044	0.008877	0.006618	0.036499	0.403634	0.828123

Appendix: 2022 complete development metrics

Model	Threshold	n	Positive rate	Brier	Log loss	PR-AUC	ROC-AUC
Frequency baseline	0	7072758	0.064804	0.060644	0.240335	0.064804	0.500000
Frequency baseline	0.5	7072758	0.035889	0.034612	0.154810	0.035889	0.500000
Frequency baseline	1	7072758	0.025229	0.024599	0.117891	0.025229	0.500000
Frequency baseline	2	7072758	0.016969	0.016684	0.086115	0.016969	0.500000
Frequency baseline	5	7072758	0.009109	0.009028	0.051979	0.009109	0.500000
Logistic regression	0	7072758	0.064804	0.038477	0.153046	0.646019	0.864880
Logistic regression	0.5	7072758	0.035889	0.020353	0.087905	0.608708	0.881650
Logistic regression	1	7072758	0.025229	0.015746	0.070586	0.553324	0.876966
Logistic regression	2	7072758	0.016969	0.012192	0.056048	0.487333	0.870590
Logistic regression	5	7072758	0.009109	0.007120	0.034610	0.401769	0.874589
Decision tree	0	7072758	0.064804	0.034893	0.155426	0.686436	0.798052
Decision tree	0.5	7072758	0.035889	0.020272	0.090266	0.613800	0.834824
Decision tree	1	7072758	0.025229	0.015371	0.070209	0.545945	0.832217
Decision tree	2	7072758	0.016969	0.011511	0.055234	0.437537	0.825002
Decision tree	5	7072758	0.009109	0.006821	0.037170	0.397890	0.831761

Appendix: 2023 complete development metrics

Model	Threshold	n	Positive rate	Brier	Log loss	PR-AUC	ROC-AUC
Frequency baseline	0	6686625	0.066713	0.062310	0.245457	0.066713	0.500000
Frequency baseline	0.5	6686625	0.037479	0.036092	0.160107	0.037479	0.500000
Frequency baseline	1	6686625	0.026297	0.025615	0.121837	0.026297	0.500000
Frequency baseline	2	6686625	0.017291	0.016996	0.087413	0.017291	0.500000
Frequency baseline	5	6686625	0.008702	0.008627	0.049975	0.008702	0.500000
Logistic regression	0	6686625	0.066713	0.038800	0.152242	0.672113	0.888477
Logistic regression	0.5	6686625	0.037479	0.020586	0.087468	0.635994	0.910461
Logistic regression	1	6686625	0.026297	0.016073	0.070502	0.571362	0.908028
Logistic regression	2	6686625	0.017291	0.012636	0.056382	0.481283	0.900575
Logistic regression	5	6686625	0.008702	0.007112	0.034088	0.384725	0.899031
Decision tree	0	6686625	0.066713	0.035137	0.155810	0.695372	0.803894
Decision tree	0.5	6686625	0.037479	0.020765	0.091492	0.625878	0.843163
Decision tree	1	6686625	0.026297	0.015974	0.071579	0.550121	0.841306
Decision tree	2	6686625	0.017291	0.011994	0.056455	0.429597	0.831717
Decision tree	5	6686625	0.008702	0.006670	0.036801	0.372486	0.831313